

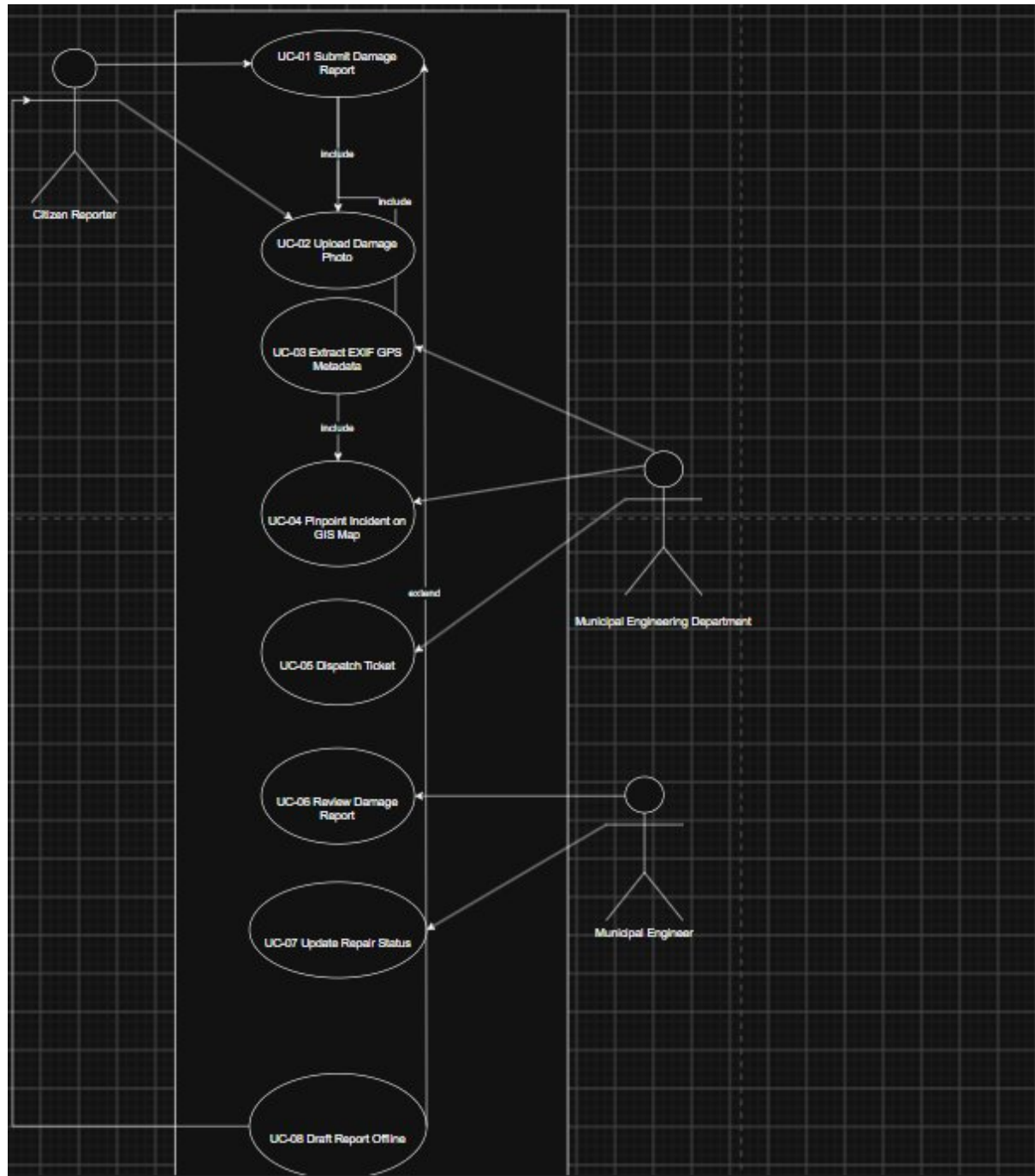
Requirements Table – Municipal Infrastructure Damage Reporting App

Problem Statement #29 | Lab 1: Requirements Engineering & UML Use-Case Modelling

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall extract EXIF GPS metadata from uploaded damage photos to automatically pinpoint the incident location on the municipal GIS map.	High	Pass: Uploaded photo with valid GPS metadata results in the incident being displayed at the corresponding GPS coordinates on the GIS map. Fail: Report is submitted without location data.	Provides accurate incident location for efficient identification and response.
FR-002	Functional	The system shall allow a Citizen Reporter to submit a damage report for road potholes, broken streetlights, or water pipeline leaks.	High	Pass: The citizen can select one of the three damage types and successfully submit the report. Fail: The system does not allow a damage type to be selected or submitted.	Core functionality for reporting municipal infrastructure damage.
FR-003	Functional	The system shall allow a Citizen Reporter to upload a photograph of the reported infrastructure damage.	High	Pass: A valid damage photograph is uploaded and attached to the corresponding report. Fail: The report cannot be submitted with a required photograph missing.	Provides visual evidence to help municipal engineers assess the reported damage.
FR-004	Functional	The system shall automatically dispatch each submitted damage report as a ticket to the relevant municipal engineering department.	High	Pass: A submitted pothole report is routed to the Road Infrastructure division, and equivalent reports are routed to their relevant department. Fail: The report remains unassigned.	Ensures that reports reach the appropriate department for action.
FR-005	Functional	The system shall allow a Municipal Engineer to review submitted damage reports and update their repair status.	Medium	Pass: A Municipal Engineer can open a submitted report and change its repair status. Fail: The engineer cannot access or update the report status.	Enables municipal staff to track and manage reported infrastructure damage.
NFR-001	Nonfunctional – Performance & Security	The reporting portal shall support offline report drafting and automatically synchronize drafted reports when mobile network connectivity resumes.	High	Pass: A report drafted while offline is saved successfully and automatically synchronized when connectivity is restored. Fail: The draft is lost or is not synchronized after connectivity resumes.	Allows citizens to prepare reports despite temporary network unavailability.
NFR-002	Nonfunctional – Performance & Security	The reporting portal shall protect submitted citizen reports and their associated location and photo data from unauthorized access.	High	Pass: Unauthorized users cannot access submitted reports, photos, or location data during security testing. Fail: An unauthorized user can retrieve or modify report information.	Protects citizen-submitted information and maintains system security.

Total: 5 Functional Requirements (FR-001 to FR-005) and 2 Non-Functional Requirements (NFR-001 to NFR-002).

UML USE-CASE DIAGRAM



USE-CASE FLOW SPECIFICATION

Municipal Infrastructure Damage Reporting App

Use Case ID: UC-01

Use Case Name: Submit Damage Report

Primary Actor: Citizen Reporter

1. PRECONDITIONS

- Citizen Reporter has access to the reporting portal.
- Citizen has a damage photo available for upload.
- Reporting application is accessible.

2. POSTCONDITIONS

- Damage report is stored in the system.
- GPS location is associated with the report.
- Incident is pinpointed on the municipal GIS map.
- Ticket is dispatched to the relevant municipal engineering department.

3. MAIN SUCCESS SCENARIO

1. Citizen Reporter opens the damage reporting portal.
2. Citizen selects "Submit Damage Report".
3. System asks the citizen to upload a damage photo.
4. Citizen uploads the damage photo.
5. System extracts EXIF GPS metadata from the photo.
6. System validates the report and GPS information.

7. System pinpoints the incident on the municipal GIS map.
8. System identifies the relevant engineering department.
9. System automatically dispatches a ticket to the department.
10. System confirms successful submission.

4. ALTERNATE FLOW – GPS LOCATION UNAVAILABLE

1. At Step 5, the system attempts to extract EXIF GPS metadata.
2. The photo does not contain valid GPS information.
3. System displays a location-unavailable message.
4. System asks the citizen to provide valid location information.
5. The report is not dispatched until valid location information is available.